

GAURIKA MAHAJAN

☎ 437-988-6918 ✉ gaurika.mahajan@mail.utoronto.ca in gaurika-mahajan 🌐 gaurikam2003
🏠 Naperville, IL | US Status: US Green Card

EDUCATION

University of Toronto (UofT) + PEY Co-op
BASc in Computer Engineering, Minor in Artificial Intelligence
AWS Certified Cloud Practitioner 📜

Toronto, ON
Sept 2022 – May 2026
Jan 2024

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS, JavaScript, TypeScript, C, C++, MATLAB, PostgreSQL
Tools & Frameworks: React, .NET, Git, MongoDB, Express, Node.js, AWS (S3, EC2, Lambda, Redshift, DynamoDB, SageMaker)
Hardware: Verilog, Arduino, Soldering, AutoCAD
Other: Excel, Oracle, Dashboard, SmartSheet, SharePoint, Microsoft Office, Google Suite

WORK EXPERIENCE

Full Stack Web Developer

May 2024 – Aug 2024

UofT Engineering Orientation Website

Toronto, ON

- Followed an **SDLC** approach to manage and redevelop a **React/Redux** website for **1,000+ engineering students**
- Implemented **Redis** to handle background subscriptions without impacting other functions, enhancing scalability **by 20%**
- Integrated payment via **Stripe API** and webhooks to process **\$150,000+** in transactions
- Created MongoDB schemas for user information; implemented indexing to **improve query performance by 40%**
- Built **Express REST API** to handle requests, including real-time leaderboard statistics and visualizations
- Used:** React, Redux, Redis, Docker, DigitalOcean, Nginx, AWS SES, MongoDB, MVC, CI/CD

Compliance Intern

May 2024 – Aug 2024

Fresenius Kabi USA

Chicago, IL

- Designed monorepo architecture to consolidate data across 7 repositories into a single pipelined workflow, **improving data retrieval speed by 95%**; automated alerts for call-to-action (CTA) responses, enhancing operational stability
- Developed data visualization pipeline, in **SmartSheet & SharePoint**, used in quarterly reports that speeds up deadline tracking, trend identification, and vendor/product reporting, **by 80%**
- Analyzed **unstructured data** (100k+ datapoints across 12 years) to create database schema, track quality agreements, and extract insights that improved product launch forecasting **from 2 days to seconds**

EXTRACURRICULARS

Software Developer

April 2024 – Present

UofT You're Next Career Network

- Built a web application with **React** and **Firestore** to manage event sign-ups, used by **3,500+ students**
- Developed a **Levenshtein** search and filter system to streamline sign-ins and navigation for career fairs, for **5,000+ users**
- Designed a user-centric interface (UI) in **Figma** to improve platform navigation and enhance student and company interactions

Web Development Director

Aug 2023 – Aug 2024

UofT Women in Science & Engineering

- Maintained and enhanced the WISE website to keep it updated, utilizing HTML, CSS, and JavaScript
- Improved the email server, ensuring efficient communication and email security **by 95%**
- Automated email forwarding and filtering using **Gmail**, improving communication efficiency and response times

PROJECTS

Skin Cancer Classifier | Python, PyTorch, TensorFlow, Google Colab, GitHub

- Developed a AI-driven CNN-based skin cancer classifier, achieving **80% test accuracy** on dermoscopic images
- Integrated unsupervised learning techniques for data augmentation and feature extraction to enhance training data quality
- Led data collection, model creation, training, and hyperparameter optimization for improved classification performance
- Produced detailed reports showcasing the model's architecture, performance metrics, & actionable insights

GIS Mapping Software | C++, Git, Valgrind

- Built and optimized a GIS application in a team of 3 using an **Agile-inspired approach** and **OpenStreetMap API**
- Implemented **A* and Dijkstra algorithms** for efficient pathfinding, leveraging **multi-threading** to cut latency by **200%**

FPGA Game Development | Assembly C

- Coded a game in **Assembly C** for an **FPGA** with visual output on a monitor, audio output, and keyboard input
- Designed smooth character movement and obstacle avoidance mechanics using **double buffering** and keyboard arrows
- Optimized **real-time performance** to handle visual rendering and audio synchronization for an engaging player experience